Subtitle Basic idea about DC to AC converter
Institute Dr B C Roy Engineering College, Durgapur
Subject Power Electronics EE 603

Inverter Lecture 10

Kingsuk Majumdar

Hello friends

Lecture 10

1. Brief recap of last lecture 09
2. Harmonics and its effects
3. Reduction techniques of harmonics
4. Signification
5. References

Harmonics in power system

- Harmonics : In an electric power system, a harmonic is a voltage or current at a multiple of the fundamental frequency of the system. [1]
- reason for harmonics:
  - non-linear load (e.g rectifiers, computer, florescent lights,)
  - saturated magnetic devices [2]
  - variable frequency drives (VFDs), etc

[1] taken https://en.wikipedia.org/wiki/Harmonics_(electrical_power)
[2] https://en.wikipedia.org/wiki/Saturation_(magnetic)

Impact of harmonics in power system

- increased heating (conductors, in the equipment etc)
- misfiring in power electronics switches say in variable speed drives
- pulsations of torque in motors

Reduction techniques of Harmonics

To kill a harmonic, you need a harmonic

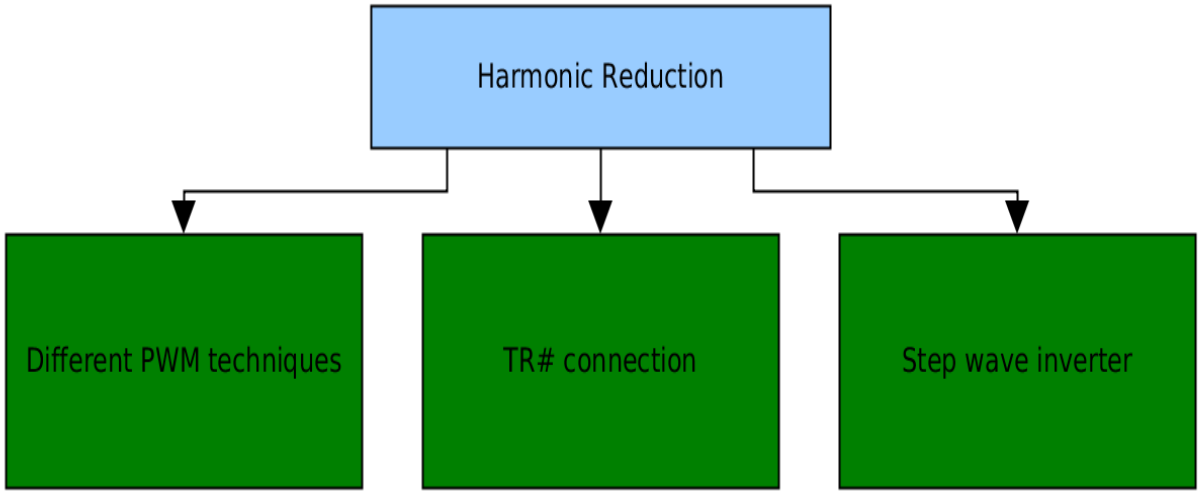


Figure: Harmonics Reduction

Different PWM techniques

already Single and multiple pulse width techniques are discussed.

- Single pulse width modulation PWM[1]
- Multiple pulse width modulation[2]

NB: If you are interested, can use MATLAB tool for
the same `https://in.mathworks.com/matlabcentral/fileexchange/56474-fourier-analysis-of-a-single-pwm-signal-a-particular-ha`

[1] `https://drive.google.com/open?id=1M9QrBihVPKWU5ZPi23Nn3Er6YGInZ-f8`
[2] `https://drive.google.com/open?id=1SW3vY1EDdJe3VdJV7EKnX6wz_UF6WIgb`

Harmonics reduction by TR# connection

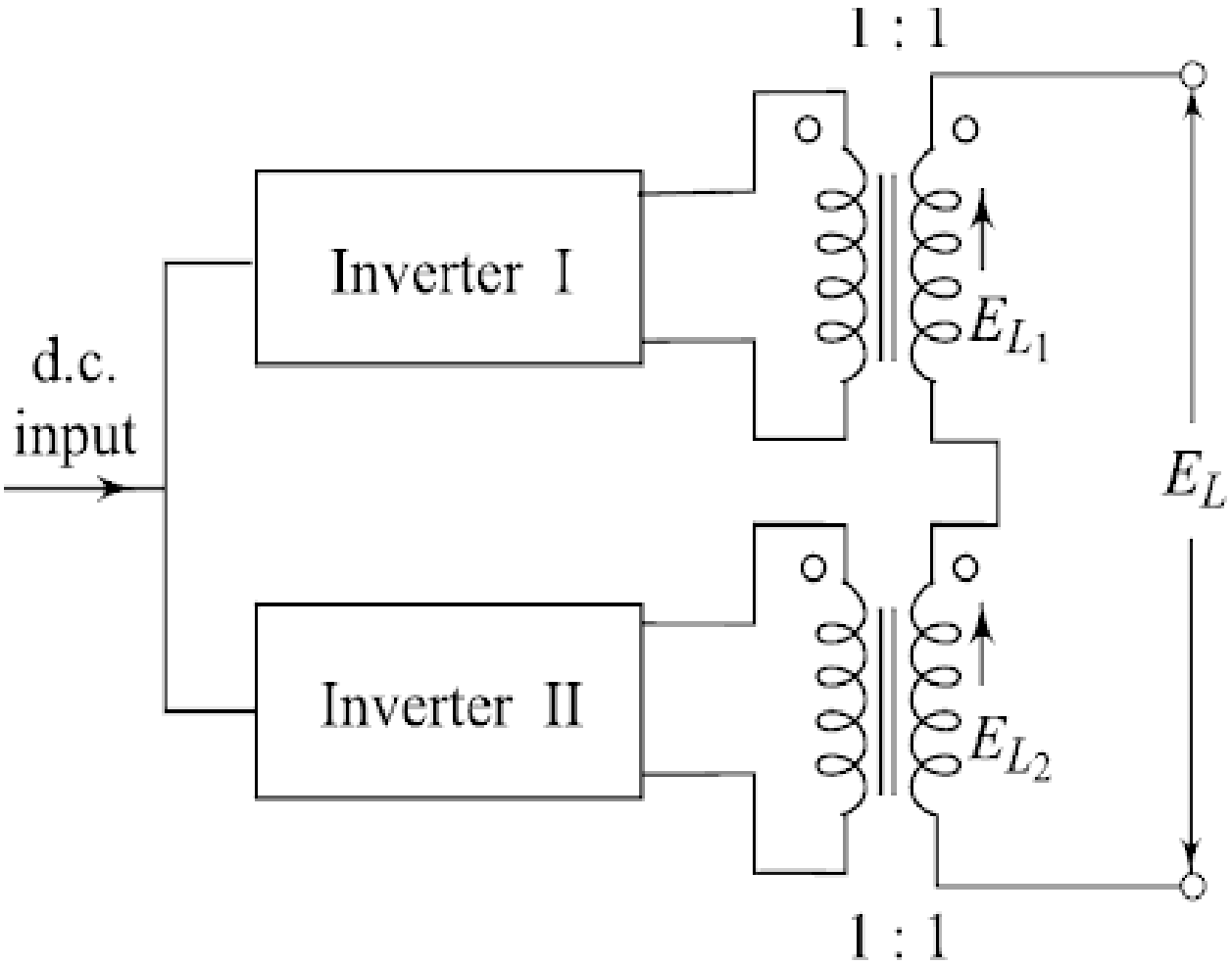


Figure: Harmonic reduction by TR# connection

NB:picture taken from google

Cont...

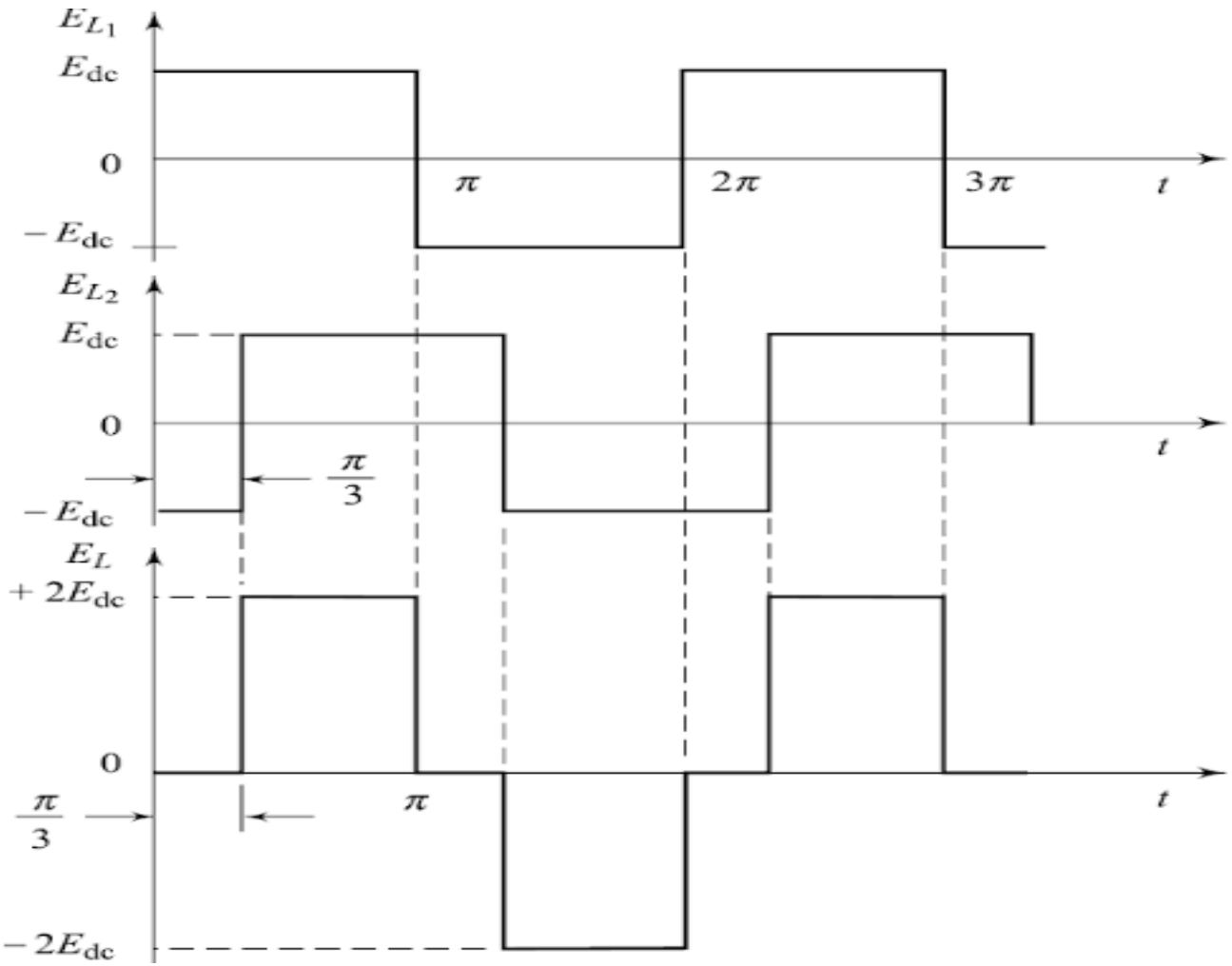


Figure: Harmonic reduction wave form by TR# connection

Cont...

Mathematics behind TR phase displacement

$$v_{0TR\#1} = A_1 \sin(\omega t) + A_3 \sin(3\omega t) + A_5 \sin(5\omega t) + ... \quad (1)$$

$$v_{0TR\#2} = A_1 \sin(\omega t - \theta_{ph}) + A_3 \sin(3\omega t - \theta_{ph}) + A_5 \sin(5\omega t - \theta_{ph}) + ... \quad (2)$$

where, θ_{ph} is the phase displacement angle.
if $\theta_{ph} = 60^0$ then

$$\sum_{i=1}^{2} v_{0TR\#i} = \sqrt{3}[A_1 \sin(\omega t - \frac{\pi}{6}) + A_5 \sin(\omega t - \frac{\pi}{6}) + ...] \quad (3)$$

From eq: 3 it is observed that output voltage does NOT contain any 3rd harmonics (NOT even any 3*n harmonics)

Harmonics reduction by step inverter

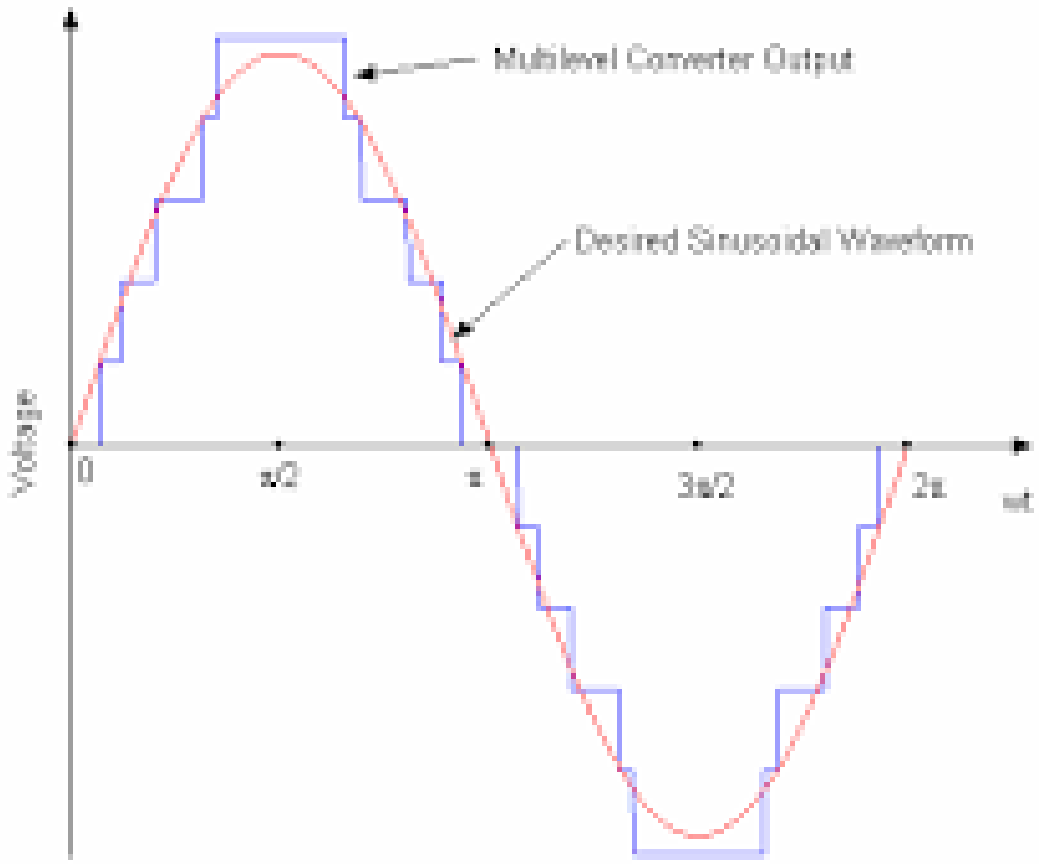


Figure: Harmonic reduction wave form by step inverter

References


N. Mohan, T. M. Undeland, and W. P. Robbins, *Power Electronics. Converters, Applications and Design*, 3rd ed. John Wiley and Sons, Inc, 2003.


M. H. Rashid, *Power electronics handbook: devices, circuits and applications*. Elsevier, 2010.


P. Bimbhra and S. Kaur, *Power electronics*. Khanna publishers, 2012, vol. 2.